Article Summary

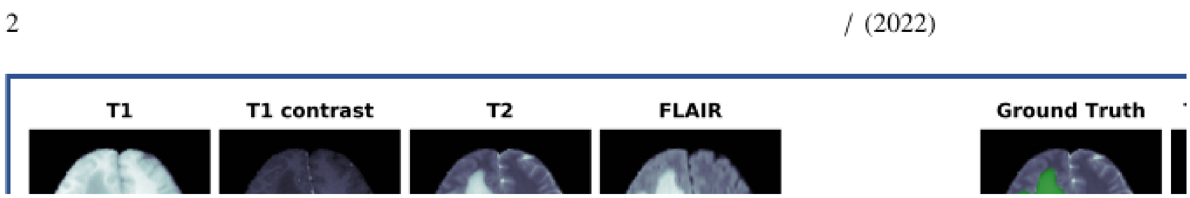
Deep and Statistical Learning in Biomedical Imaging: State of the Art in 3D MRI Brain Tumor Segmentation

K. Ruwani M. Fernando and Chris P. Tsokos (2022)

1. Purpose and Scope

This article is a broad methodological review of 3D MRI brain-tumor segmentation. Rather than treating classical statistics and deep learning as competing traditions, Fernando and Tsokos organize the field around three interacting paradigms: **model-driven statistical methods**, **data-driven deep learning**, and **hybrid/probabilistic deep learning**. Their central thesis is that combining domain-informed statistical modeling with representation-learning capacity from deep networks is especially promising for robust automated neuro-oncologic imaging.

The clinical motivation is precise volumetric delineation of heterogeneous glioma tissue. MRI provides complementary T1, contrast-enhanced T1, T2, and FLAIR information, while segmentation separates tumor subregions such as enhancing tumor, necrotic/non-enhancing core, and peritumoral edema.



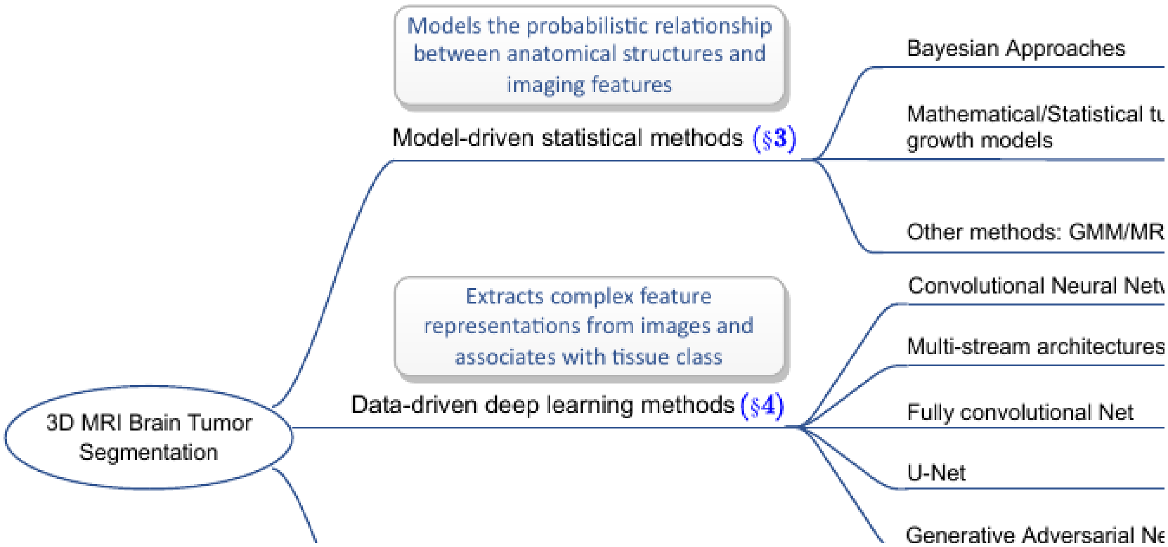
Article Fig. 1 (cropped from p. 2). Multimodal glioblastoma MRI and the corresponding voxel-wise ground-truth tumor segmentation.

2. Why Segmentation Matters Clinically

Accurate segmentation supports neurosurgical planning and navigation, tumor-volume measurement, assessment of resection extent, visualization of complex pathology, diagnosis, risk assessment, and treatment planning. The review emphasizes that manual contouring remains time-consuming and subject to inter-observer variability, motivating automated, reproducible 3D methods.

MRI preprocessing commonly includes **registration** to align images, **skull stripping** to remove non-brain tissues, and correction of **intensity inhomogeneity/bias field**. The authors also note that preprocessing choices can themselves affect pathological signals.

3. Taxonomy of the Field



Article Fig. 2 (cropped from p. 4). The review's core taxonomy: statistical, deep-learning, and hybrid methods, plus major applications of integrated approaches.

4. Model-Driven Statistical Segmentation

Bayesian modeling relates tissue labels to image features through priors, likelihoods, and posterior inference. Its strength is the ability to encode anatomical or spatial knowledge and provide probabilistic interpretations; its weakness is sensitivity to prior specification and difficult inference in high-dimensional images.

Biophysical and mathematical tumor-growth models explicitly model tumor evolution and cell-density patterns. Frameworks such as GLISTR combine growth modeling with registration and segmentation, illustrating how mechanistic oncology can inform image analysis.

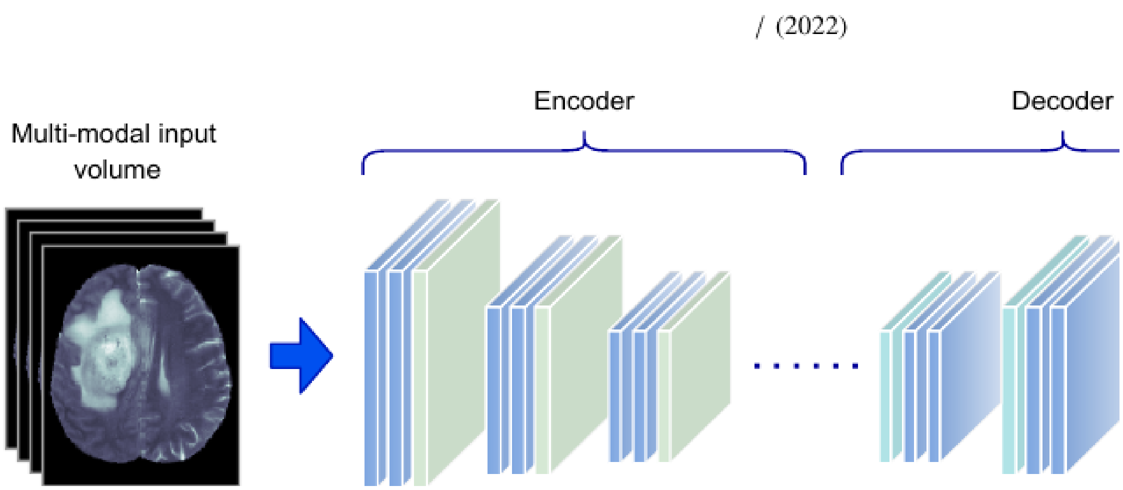
Gaussian mixture models (GMMs) model intensity distributions but often make unrealistic spatial-independence assumptions. **Markov random fields (MRFs)** address this by modeling neighborhood dependencies between voxels, though posterior computation can be expensive. **Expectation-Maximization (EM)** is frequently paired with GMM/HMRF models to estimate latent labels and unknown parameters, but can converge slowly or to local optima.

The authors' overall assessment is that classical statistical methods are relatively interpretable and can work with less data, but require domain expertise, explicit probabilistic assumptions, and potentially expensive inference.

5. Data-Driven Deep Learning

CNNs automatically learn hierarchical image features. Patch-wise CNN segmentation classifies the center voxel of local image patches, but overlapping patches create redundant computation and limited patch sizes can restrict global context. **3D CNNs** extend convolution to volumetric kernels and can exploit information across adjacent slices.

Multi-path architectures fuse information from different views or scales. **Fully convolutional networks (FCNs)** replace fully connected classification layers with convolutional prediction and use upsampling to restore spatial resolution. **U-Net** became dominant by linking contracting and expanding paths through skip connections, enabling precise localization while retaining high-level context. 3D U-Net, V-Net, cascaded U-Net and nnU-Net are highlighted as important volumetric variants.



Article Fig. 4 (cropped from p. 9). SegNet-like encoder-decoder segmentation: downsampling learns representations and decoder upsampling restores a full-resolution segmentation.

GANs add adversarial learning: a segmentation network generates masks while a discriminator attempts to distinguish predicted masks from expert ground truth. This can encourage anatomically plausible outputs and has also been explored for domain adaptation.

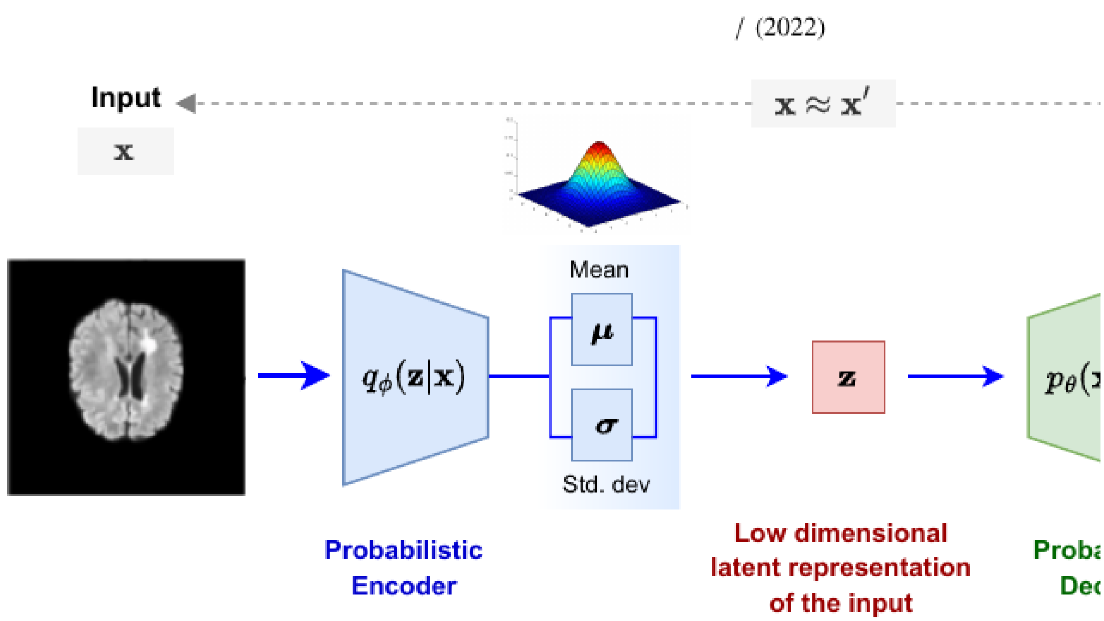
6. Evolution of BRATS Winning Systems

The review summarizes a clear progression in BRATS challenge winners: ensembles of 3D CNN/FCN/U-Net models (2017), encoder-decoder CNNs with VAE regularization (2018), cascaded U-Nets (2019), modified nnU-Net (2020), and an extended nnU-Net (2021). Reported test-set Dice scores improved substantially by BRATS 2021, especially for whole tumor.

Challenge	Architecture	WT Dice	ET Dice	TC Dice
2017	Ensemble: DeepMedic + 3D U-Net + FCN	0.886	0.729	0.785
2018	Encoder-decoder CNN + VAE	0.8839	0.7664	0.8154
2019	Two-stage cascaded U-Net	0.8880	0.8327	0.8370
2020	Modified nnU-Net	0.8895	0.8203	0.8506
2021	Extended nnU-Net	0.9319	0.8835	0.8878

7. Probabilistic Deep Learning and Hybrid Methods

The review treats probabilistic deep learning as the bridge between the two traditions. **Variational autoencoders (VAEs)** learn distributions in a latent space rather than only deterministic representations. In brain imaging they can model normal anatomy, detect abnormalities as deviations, and regularize segmentation networks. **Bayesian neural networks (BNNs)** place probability distributions over model parameters or functions, enabling probabilistic predictions and explicit uncertainty estimation.



Article Figs. 6-7 (cropped from p. 13). Top: probabilistic VAE reconstruction through a latent distribution. Bottom: a deep segmentation network followed by fully connected CRF boundary refinement.

Hybrid approaches are grouped around four important applications: **scarce annotations** (e.g., probabilistic atlases, EM and semi-supervision), **segmentation refinement** (MRF/CRF post-processing or integrated CRF networks), **uncertainty quantification** (Bayesian networks and evidential approaches), and **generalizability** across institutions, scanners and modalities.

8. Datasets and Evaluation

The paper highlights BRATS, FeTS and the Medical Segmentation Decathlon as important public benchmarks. BRATS 2021 contained 2,040 cases (1,251 training, 219 validation and 570 testing) with T1, T1ce, T2 and FLAIR MRI and expert tumor-subregion annotations. FeTS emphasizes federated learning and domain generalization without centralizing institutional data. The MSD brain-tumor task contains multimodal glioma MRI from 750 patients.

Evaluation centers on overlap and boundary measures. **Dice coefficient** is the most widely used overlap metric, with 1 indicating perfect overlap. The **Jaccard index** measures intersection over union. **Hausdorff distance** evaluates disagreement between segmentation boundaries. Sensitivity and specificity are also used, although the authors note limitations for differently sized structures.

9. Major Challenges

The review identifies several obstacles to dependable clinical deployment. Brain lesions have subtle boundaries, low contrast, intensity inhomogeneity, noise and partial-volume effects. **Class imbalance** is particularly severe because tumor may constitute only about 15% of the brain and individual tumor subregions are smaller still. **Scarce expert annotations** constrain supervised learning. Finally, full 3D models impose substantial **memory and computational costs**, creating a trade-off between volumetric context and practical resource requirements.

10. Future Directions

The authors frame the next phase around trustworthiness rather than raw benchmark accuracy alone. Priorities include learning features that remain stable across scanners and acquisition settings; quantifying uncertainty so a model can signal when it is unsure; detecting automated-system failure when no expert ground truth is available; and ultimately moving beyond human-level performance while acknowledging uncertainty and disagreement in expert annotations.

11. Overall Conclusion

Fernando and Tsokos conclude that deep learning has become a powerful mechanism for automated volumetric segmentation, but that **integrating statistical methodology into deep networks can improve biomedical image segmentation**. They therefore view the intersection of statistical modeling and deep learning as a more effective direction than relying exclusively on either paradigm.

The review's most important conceptual message is that clinically useful medical AI needs more than a high Dice score. A robust system must combine rich representation learning with probabilistic reasoning, anatomical or mechanistic knowledge, uncertainty awareness, generalization across clinical environments, and reliable failure detection.

Source: Fernando KRM, Tsokos CP. *Deep and Statistical Learning in Biomedical Imaging: State of the Art in 3D MRI Brain Tumor Segmentation*. 2022.